Optimal Low-Thrust Transfers under the Circular Restricted Three-Body Problem

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This work studies the process in determining the minimum-fuel optimal transfer trajectory between two halo orbits under the Circular Restricted Three-Body Problem with a low-thrust capable satellite. First the shooting method is explored with the co-state vector initialized from previous works, and then the steepest descent philosophy is used to determine trajectories without knowledge of co-states that converge to optimal solutions. A continuation method is considered to transition minimum-energy solutions to minimum-fuel solutions and vice-versa.

I. Problem Context and Setup

A. Circular Restricted Three-Body Problem

The intention of this work is to study methods of determining control profiles for satellites in the cislunar region. The cislunar domain is defined as the area in space where a space object (SO) is under the gravitational influence of primary bodies, the Earth and Moon. The three-body problem, as it is commonly referred to, is highly non-linear with no analytical realizations. The Circular Restricted Three-Body Problem (CR3BP) is a simplified model that assumes the third body is of negligible mass and size compared to the two, much larger, primary bodies and its motion is only governed by the gravitational forces of the two larger bodies. The CR3BP further assumes the two primary bodies, the Earth and the Moon, are rotating in a circular orbit about their common center of mass, the barycenter, at a constant angular rate. It is common practice to express the position of the SO in a reference frame that rotates with the primary bodies, referred to as the synodic frame. The origin of the synodic frame is placed at the barycenter of the three-body system, the $\mathbf{e_x}$ is defined along the axis of the primary bodies in the direction towards the smaller primary body, the Moon, the $\mathbf{e_z}$ direction is defined along the axis of rotation of the primary bodies about their barycenter, and the $\mathbf{e_y}$ direction completes the right-handed coordinate system. A diagram of the synodic frame is shown in Fig. 1. When considering low-thrust implementations the state vector is expressed

$$\mathbf{x} = \begin{bmatrix} \mathbf{r} \\ \mathbf{v} \\ m \end{bmatrix} \tag{1}$$

where $\mathbf{r} = [x \;\; y \;\; z]^T$ is the satellites position in the $\mathbf{e_x}$, $\mathbf{e_y}$, and $\mathbf{e_z}$ directions of the synodic frame, respectfully, $\mathbf{v} = [\dot{x} \;\; \dot{y} \;\; \dot{z}]^T$ is the satellites velocities in the same directions, and m is the mass of the satellite (note the T superscript denotes the transpose operator). The state dynamics under low-thrust are formulated as [1],

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\alpha}, u) = \begin{bmatrix} \dot{\mathbf{r}} \\ \dot{\mathbf{v}} \\ \dot{m} \end{bmatrix} = \begin{bmatrix} \mathbf{v} \\ \mathbf{g}(\mathbf{r}) + \mathbf{h}(\mathbf{v}) + uT_{max}\boldsymbol{\alpha}/m \\ -uT_{max}/c \end{bmatrix} \tag{2}$$

with T_{max} being the maximum thrust allowable, c is the propellant exhaust velocity determined by $c = I_{sp}g_0$ where I_{sp} is the specific impulse of the satellite and $g_0 = 9.81$ m/s^2 is the acceleration due to gravity near Earth. The control variables are $u \in [0 \;\; 1]$, which represents the thrust throttle, determining the magnitude of force provided, and $\boldsymbol{\alpha} \in \mathbb{R}^3$ is a unit vector that points in the direction of the thrust. The functions $\mathbf{g}(\mathbf{r})$ and $\mathbf{h}(\mathbf{v})$ are given by,

$$\mathbf{g}(\mathbf{r}) = \begin{bmatrix} x - (1-\mu)\left(\frac{x+\mu}{r_1^3}\right) - \mu\left(\frac{x+\mu-1}{r_2^3}\right) \\ y - \frac{(1-\mu)y}{r_1^3} - \frac{\mu y}{r_2^3} \\ -\frac{(1-\mu)z}{r_1^3} - \frac{\mu z}{r_2^3} \end{bmatrix}, \quad \mathbf{h}(\mathbf{v}) = \begin{bmatrix} 2\dot{x} \\ -2\dot{y} \\ 0 \end{bmatrix} \tag{3}$$

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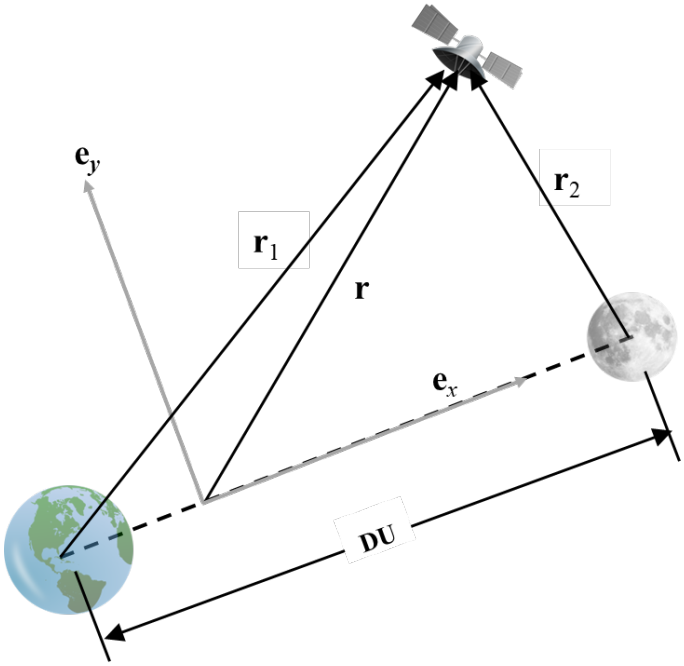


Fig. 1 Synodic Reference Frame Definition

where,

$$r_1 \equiv \sqrt{(x+\mu)^2 + y^2 + z^2} \tag{4a}$$

$$r_2 \equiv \sqrt{(x+\mu-1)^2 + y^2 + z^2} \tag{4b}$$

where r_1 and r_2 are the distances of the spacecraft from the Earth and the Moon, respectively. Here μ is the non-dimensional mass ratio between the mass of the Moon to its summed with the mass of the Earth, given by

$$\mu \equiv \frac{m_{\leftmoon}}{m_{\oplus} + m_{\leftmoon}} \tag{5}$$

Note that the equations of motion in the CR3BP are normalized such that the positional quantities, x, y, and z have non-dimensional units of distance, (DU), while the velocity variables are expressed as $^{DU}/_{TU}$, where TU is a non-dimensional time unit. Dimensional quantities can be recovered using simple scaling:

$$d' = d\,L_c \tag{6a}$$

$$t' = t\,T_c \tag{6b}$$

where d and t represent non-dimensional values of distance and time and d' and t' represent the dimensioned values. The characteristic length, L_c, is defined as the average distance between the primary bodies, while the characteristic time, T_C, is defined as the time for the bodies to complete one revolution about their barycenter.

II. Formulation of the Control Problem

A. Minimum-Fuel Trajectory Optimization using Energy-to-Fuel Homotopy

For the minimum-fuel problem the optimal trajectory seeks to minimize the cost function,

$$J_{MF} = \frac{T_{max}}{c} \int_{t_0}^{t_f} u\, dt \tag{7}$$

where t_0 and t_f represent the initial and final times of the transfer. The initial and final states, $\mathbf{x}_0$ and $\mathbf{x}_f$ respectively, are fixed, as well as initial and final times. Notably, the minimum-fuel trajectories are characterized by a *bang-bang* control profile, in which the throttle factor, u, is either 1 or 0. This results in discontinuities over the control profile, which often causes trouble for numeric solvers evaluating such dynamics. Luckily, a solution that mitigates these effects was presented in Ref. [2] that uses an energy-to-fuel homotopy continuation that gradually transitions solutions of the minimum-energy problem, which do not have these discontinuities, to the minimum-fuel problem. In other words, the control profile is initially solved such that it is continuous, and little by little the discontinuities are enforced until the *bang-bang* profile is fully realized. The cost function related to the minimum-energy problem is stated as,

$$J_{ME} = \frac{T_{max}}{c} \int_{t_0}^{t_f} \left(u - \epsilon u(1-u)\right)\ dt \tag{8}$$

where $\epsilon \in [0\ \ 1]$ is the perturbing parameter that smooths the control profile. When $\epsilon = 1$ the resulting optimal trajectory corresponds to the minimum-energy problem, and when $\epsilon = 0$, the optimal trajectory corresponds to the minimum-fuel problem. The minimum-energy trajectory is solved and then the value of ϵ is decreased slightly, and that trajectory is solved for using the previous solution as an initial guess. This process is repeated until $\epsilon = 0$.

To solve for such trajectories, we first formulate the Two-Point Boundary Value Problem (TPBVP). We form the Hamiltonian, $\mathcal{H}$ as,

$$\begin{aligned}\mathcal{H} &= \frac{T_{max}}{c}\left(u - \epsilon u(1-u)\right) + \boldsymbol{\lambda}^T \mathbf{f}(\mathbf{x}, \boldsymbol{\alpha}, u) \\ &= \frac{T_{max}}{c}\left(u - \epsilon u(1-u)\right) + \boldsymbol{\lambda}_r^T\left[\mathbf{v}\right] + \boldsymbol{\lambda}_v^T\left[\mathbf{g}(\mathbf{r}) + \mathbf{h}(\mathbf{v}) + uT_{max}\boldsymbol{\alpha}/m\right] - \lambda_m \frac{uT_{max}}{c}\end{aligned} \tag{9}$$

where $\boldsymbol{\lambda} = [\boldsymbol{\lambda}_r\ \ \boldsymbol{\lambda}_v\ \ \lambda_m]^T$ is the co-state vector with $\boldsymbol{\lambda}_r = [\lambda_x\ \ \lambda_y\ \ \lambda_z]^T$ and $\boldsymbol{\lambda}_v = [\lambda_{\dot{x}}\ \ \lambda_{\dot{y}}\ \ \lambda_{\dot{z}}]^T$ where each component represents the co-state of each state, denoted in the subscript. The co-state evolution over time is

$$\dot{\boldsymbol{\lambda}} = \begin{bmatrix}\dot{\boldsymbol{\lambda}}_r \\ \dot{\boldsymbol{\lambda}}_v \\ \dot{\lambda}_m\end{bmatrix} = -\frac{\partial \mathcal{H}}{\partial \mathbf{x}} = \begin{bmatrix}-G^T\boldsymbol{\lambda}_v \\ -\boldsymbol{\lambda}_r - H^T\boldsymbol{\lambda}_v \\ \frac{uT_{max}}{c}\boldsymbol{\lambda}_v^T\boldsymbol{\alpha}\end{bmatrix} \tag{10}$$

where the matrices G and H are defined as,

$$G = \frac{\partial \mathbf{g}(\mathbf{r})}{\partial \mathbf{r}} \tag{11a}$$

$$H = \frac{\partial \mathbf{h}(\mathbf{v})}{\partial \mathbf{v}} \tag{11b}$$

where the analytical derivatives for G necessary are

$$G_{1,1} = 1 - \frac{(1-\mu)}{r_1^3} - \frac{\mu}{r_2^3} + \frac{3(1-\mu)(x+\mu)^2}{r_1^5} + \frac{3\mu(x+\mu-1)^2}{r_2^5} \tag{12a}$$

$$G_{1,2} = G_{2,1} = \frac{3(1-\mu)(x+\mu)y}{r_1^5} + \frac{3\mu(x+\mu-1)y}{r_2^5} \tag{12b}$$

$$G_{1,3} = G_{3,1} = \frac{3(1-\mu)(x+\mu)z}{r_1^5} + \frac{3\mu(x+\mu-1)z}{r_2^5} \tag{12c}$$

$$G_{2,2} = 1 - \frac{(1-\mu)}{r_1^3} - \frac{\mu}{r_2^3} + \frac{3(1-\mu)y^2}{r_1^5} + \frac{3\mu y^2}{r_2^5} \tag{12d}$$

$$G_{2,3} = G_{3,2}\frac{3(1-\mu)yz}{r_1^5} + \frac{3\mu yz}{r_2^5} \tag{12e}$$

$$G_{3,3} = -\frac{(1-\mu)}{r_1^3} - \frac{\mu}{r_2^3} + \frac{3(1-\mu)z^2}{r_1^5} + \frac{3\mu z^2}{r_2^5} \tag{12f}$$

and H is simply,

$$H = \begin{bmatrix} 0 & 2 & 0 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{13}$$

The boundary conditions on the co-states relating to the position and velocity are free due to the fixed nature of the boundary conditions on the actual states corresponding to the position and velocity. The final mass, however, is free and its corresponding co-state is constrained to be zero at the final time, i.e.,

$$\lambda_m(t_f) = 0 \tag{14}$$

Pontryagin's Maximum Principle (PMP) is used to derive conditions on the control variables u and $\boldsymbol{\alpha}$. Minimizing the Hamiltonian and assuming that $uT_{max}/c \geqslant 0$ results in the optimal direction of thrust,

$$\boldsymbol{\alpha}^* = -\frac{\boldsymbol{\lambda}_v}{\lambda_v} \tag{15}$$

where the superscript $(\cdot)^*$ indicates an optimal value. The Hamiltonian can then be rewritten as,

$$\mathcal{H} = \boldsymbol{\lambda}_r^T [\mathbf{v}] + \boldsymbol{\lambda}_v^T [\mathbf{g}(\mathbf{r}) + \mathbf{h}(\mathbf{v})] + \frac{uT_{max}}{c} (S - \epsilon + \epsilon u) \tag{16}$$

where S is a switching function defined as,

$$S = -\lambda_m \frac{c}{m} - \lambda_m + 1 \tag{17}$$

where the first time derivative is expressed as,

$$\dot{S} = \frac{c}{m} \left(\boldsymbol{\lambda}_r + H^T \boldsymbol{\lambda}_v\right)^T \frac{\boldsymbol{\lambda}_v}{\lambda_v} \tag{18}$$

From this, and another application of the PMP, the optimal throttle factor, u^* is determined using,

$$u^* = \begin{cases} 0 & S > \epsilon \\ \frac{\epsilon - S}{2\epsilon} & -\epsilon \leqslant S \leqslant \epsilon \\ 1 & S < -\epsilon \end{cases} \tag{19}$$

Note that the *bang-bang* profile of the the throttle is a natural occurrence of this formulation when $\epsilon = 0$, as well as the discontinuity that results when $S = 0$. Now that the control variables are written in terms of the co-states, we can form an augmented state vector, $\mathbf{y} = [\mathbf{x} \;\; \boldsymbol{\lambda}]^T$ subject to the dynamics,

$$\dot{\mathbf{y}} = \begin{bmatrix} \dot{\mathbf{r}} \\ \dot{\mathbf{v}} \\ \dot{m} \\ \dot{\boldsymbol{\lambda}}_r \\ \dot{\boldsymbol{\lambda}}_v \\ \dot{\lambda}_m \end{bmatrix} = \begin{bmatrix} \mathbf{v} \\ \mathbf{g}(\mathbf{r}) + \mathbf{h}(\mathbf{v}) + uT_{max}\boldsymbol{\alpha}/m \\ -uT_{max}/c \\ -G^T \boldsymbol{\lambda}_v \\ -\boldsymbol{\lambda}_r - H^T \boldsymbol{\lambda}_v \\ -\lambda_m uT_{max}/m^2 \end{bmatrix} \tag{20}$$

B. Shooting Method

As the control variables were eliminated (or rather can be represented as functions of the co-states), the shooting method is an obvious choice for determining an optimal trajectory, and coincidentally has been successful to varying degrees across the works mentioned thus far.

The shooting method is characterized by guessing an initial condition for the unknown co-states and propagating them forward in time and seeing if the final conditions for optimality are met. In the ,almost certain, event this initial

guess does not meet the final conditions for optimality, the initial guess is changed by leveraging the difference in the propagated final states and co-states to that of the desired final states and co-states. We define these errors as such,

$$\Delta\mathbf{x}(t) = \mathbf{x}^*(t) - \mathbf{x}(t) \tag{21a}$$
$$\Delta\boldsymbol{\lambda}(t) = \boldsymbol{\lambda}^*(t) - \boldsymbol{\lambda}(t) \tag{21b}$$

The final errors can be related to the initial errors by the State Transition Matrix (STM), $\boldsymbol{\Phi}(t, t_0)$ through the relationship,

$$\begin{bmatrix} \Delta\mathbf{x}(t_f) \\ \Delta\boldsymbol{\lambda}(t_f) \end{bmatrix} = \boldsymbol{\Phi}(t_f, t_0) \begin{bmatrix} \Delta\mathbf{x}(t_0) \\ \Delta\boldsymbol{\lambda}(t_0) \end{bmatrix} \tag{22}$$

where the STM can be propagated numerically using the dynamic relationship,

$$\dot{\boldsymbol{\Phi}}(t,t_0) = F\,\boldsymbol{\Phi}(t, t_0) \tag{23}$$

where F is the Jacobian matrix of the augmented state vector in Eq. (20) and the STM is initialized with $\boldsymbol{\Phi}(t_0, t_0) = I_{14\times 14}$, with I of course being the identity matrix. The Jacobian matrix F is found for two scenarios. The first, $F^{(1)}$, for when $u^* = 0$ or $u^* = 1$, and the second, $F^{(2)}$ for when $u^* = (\epsilon - S)/2\epsilon$, these expressions are expressed as [1],

$$F^{(1)} = \left[\begin{array}{cccc:cc} \mathbf{0} & I & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ G & H & \frac{\lambda_v}{\lambda_v}\frac{uT_{max}}{m^2} & \mathbf{0} & -\frac{uT_{max}}{m}\left(\frac{I}{\lambda_v} - \frac{\lambda_v {\lambda_v}^T}{\lambda_v^3}\right) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & 0 & \mathbf{0} & \mathbf{0} & 0 \\ \hdashline -\frac{\partial\left(\bar{G}^T\bar{\lambda}_v\right)}{\partial\mathbf{r}} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -G^T & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & -I & -H^T & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \frac{2\lambda_v uT_{max}}{m^3} & \mathbf{0} & -\frac{\lambda_v^T}{\lambda_v}\frac{uTmax}{m^2} & 0 \end{array}\right] \tag{24}$$

$$F^{(2)} = \left[\begin{array}{cccc:cc} \mathbf{0} & I & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ G & H & \boldsymbol{\Omega}_1 & \mathbf{0} & \boldsymbol{\Omega}_2 & \boldsymbol{\Omega}_3 \\ \mathbf{0} & \mathbf{0} & \Omega_4 & \mathbf{0} & \boldsymbol{\Omega}_5 & \Omega_6 \\ \hdashline -\frac{\partial\left(\bar{G}^T\bar{\lambda}_v\right)}{\partial\mathbf{r}} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -G^T & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & -I & -H^T & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \Omega_7 & \mathbf{0} & \boldsymbol{\Omega}_8 & \Omega_9 \end{array}\right] \tag{25}$$

where

$$\boldsymbol{\Omega}_1 = \frac{\boldsymbol{\lambda}_v}{\lambda_v}\frac{uT_{max}}{m^2} + \frac{\boldsymbol{\lambda}_v cT_{max}}{2\epsilon m^3} \qquad \boldsymbol{\Omega}_2 = \frac{\boldsymbol{\lambda}_v {\boldsymbol{\lambda}_v}^T}{\lambda_v^2}\frac{cT_{max}}{2\epsilon m^2} - \frac{uT_{max}}{m}\left(\frac{I}{\lambda_v} - \frac{\boldsymbol{\lambda}_v {\boldsymbol{\lambda}_v}^T}{\lambda_v^3}\right) \qquad \boldsymbol{\Omega}_3 = -\frac{\boldsymbol{\lambda}_v}{\lambda_v}\frac{uT_{max}}{2\epsilon m}$$

$$\Omega_4 = \frac{\lambda_v T_{max}}{2\epsilon m^2} \qquad \boldsymbol{\Omega}_5 = -\frac{\boldsymbol{\lambda}_v^T}{\lambda_v}\frac{T_{max}}{2\epsilon m} \qquad \Omega_6 = -\frac{T_{max}}{2\epsilon c}$$

$$\Omega_7 = \frac{2\lambda_v uT_{max}}{m^3} + \frac{\lambda_v^2 cT_{max}}{2\epsilon m^4} \qquad \boldsymbol{\Omega}_8 = -\frac{\boldsymbol{\lambda}_v^T}{\lambda_v}\frac{uTmax}{m^2} - \frac{\boldsymbol{\lambda}_v^T cT_{max}}{2\epsilon m^3} \qquad \Omega_9 = -\frac{\lambda_v T_{max}}{2\epsilon m^2}$$

At this point I would like to acknowledge Ref. [1] for providing Eqs. (24)-(25), but I would also like to state my appreciation for not providing the $\partial\left(G^T\boldsymbol{\lambda}_v\right)/\partial\mathbf{r}$ terms because taking those derivatives, while a seemingly annoying and tedious task, I did end up enjoy doing them. To see these derivatives fully written, refer to Appendix A.

The mapping of states using $\boldsymbol{\Phi}(t, t_0)$ are only valid for a continuous orbit. If at any time t_j there is a discontinuity (i.e. when the switching function switches) a corrective term in the STM is included to account for such a discontinuity, $\boldsymbol{\Psi}(t_j)$. This term is calculated as [3],

$$\boldsymbol{\Psi}(t_j) = I_{14\times 14} + \left(\dot{\mathbf{y}}|_{t_j^+} - \dot{\mathbf{y}}|_{t_j^-}\right)\left(\frac{\partial S}{\partial\mathbf{y}}\frac{1}{\dot{S}}\right)\Bigg|_{t_j^-} \tag{26}$$

Table 1 CR3BP Parameters

	Value	Units
μ	$1.21506038 \times 10^{-2}$	N/A
L_c	384400	km
T_c	375162.997	s

where t_j^- and t_j^+ are the times immediately before and after the switch occurs. I use a buffer time of 1×10^{-10} TU. As multiple discontinuities are to be expected across a trajectory, the total STM can be expressed as a composite of STMs [3],

$$\mathbf{\Phi}(t_f, t_0) = \mathbf{\Phi}(t_f, t_N^+)\mathbf{\Psi}(t_N)\mathbf{\Phi}(t_N^-, t_{N-1}^+)\mathbf{\Psi}(t_{N-1})\ldots\mathbf{\Phi}(t_2, t_1^+)\mathbf{\Psi}(t_1)\mathbf{\Phi}(t_1^-, t_0) \tag{27}$$

where $t_1,\ t_2,\ \ldots,\ t_{N-1},\ t_N$ represent the times of N number of discontinuities. Plenty of papers provide information on switching detection, but I found using MATLAB's built-in odeEvent functionality to be extremely accurate and easy to implement. I simply propagate the states and co-states forward in time with the current initial condition guess and extract the times of the switching from that function call in order to create the STM afterwords. This is certainly not the most efficient way, since propagating the STM includes propagating the states and co-states for a second time, but modern computers make it easy to ignore such trivial things.

C. Some Initial Results

To begin, we first seek to reproduce results from another paper. Reference [4] references Reference [5] which both use the parameters listed in Table 1. The first scenario evaluated is a transfer from an L_2 halo orbit to an L_1 halo orbit. Note these halo orbits are not "mirrored" versions of each other so we cannot simply use heteroclinic transfers (that's like the whole point). The initial mass of the satellite is $m_i = 2000$ kg, with a maximum thrust $T_{max} = 1.5$ N, and a specific impulse of $I_{sp} = 2000$ s. The duration of the transfer is set to 12.7 days, and the initial state, $\mathbf{x_0} = \begin{bmatrix}\mathbf{r}_0^T & \mathbf{v}_0^T\end{bmatrix}^T$ and the final state, $\mathbf{x}_f = \begin{bmatrix}\mathbf{r}_f^T & \mathbf{v}_f^T\end{bmatrix}^T$ are

$$\mathbf{r}_0 = \begin{bmatrix}1.1599795702248494 & 0.009720428035815552 & -0.12401864915284157\end{bmatrix}^T\ DU \tag{28a}$$

$$\mathbf{v}_0 = \begin{bmatrix}0.008477705130550553 & -0.20786307954141953 & -0.010841912833115475\end{bmatrix}^T\ VU \tag{28b}$$

$$\mathbf{r}_f = \begin{bmatrix}0.8484736688482315 & 0.00506488863463682 & 0.17343680487577373\end{bmatrix}^T\ DU \tag{28c}$$

$$\mathbf{v}_f = \begin{bmatrix}0.005241131023638693 & 0.26343491250951045 & -0.008541420325316247\end{bmatrix}^T\ VU \tag{28d}$$

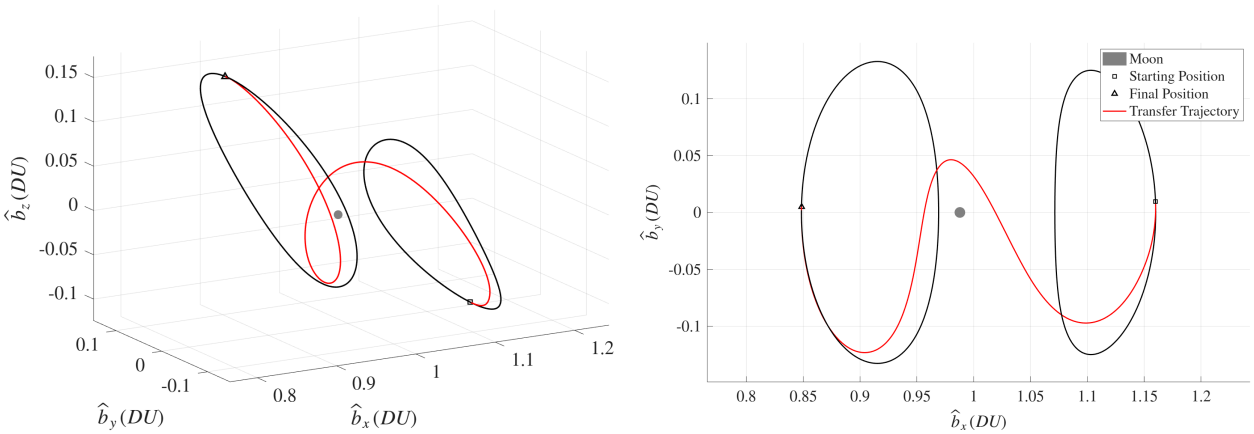
The period of the L_2 orbit is 14.2 days and the period of the L_1 orbit is 11.2 days To ensure the simulation is properly set up, the solution found in Ref. [4] that corresponds to the minimum-fuel transfer presented is used as an initial guess.

$$\boldsymbol{\lambda}_r(t_0) = \begin{bmatrix}0.12603 & -0.07665 & -0.05635\end{bmatrix}^T \tag{29a}$$

$$\boldsymbol{\lambda}_v(t_0) = \begin{bmatrix}0.03999 & -0.00518 & -0.06410\end{bmatrix}^T \tag{29b}$$

$$\lambda_m(t_0) = 0.02236 \tag{29c}$$

It is lucky that the initial co-state vector given is only listed to a few decimal places, as it results in the the final propagated state to be about 900 km away from where it should, so we get to actually test the shooting method implementation without worrying about whether the initial guess is so bad which causes the trajectories to diverge (something I've spent days circling before realizing when working on another problem). After a few days debugging various issues, the correct optimal trajectory was achieved as shown in Figure 2. The control profile for the throttle and switching curve is shown in Figure 3. The TPBVP was solved in six iterations with the difference in final position being about 4.03 cm and the discrepancies in the final velocity end up being on the order of 10^{-4} mm/s in magnitude. As in Ref [4] the final



(a) Three-dimensional view **(b) Planar projection**

Fig. 2 Minimum-Fuel trajectory

change in mass is $\Delta m = 35.34$ kg. The initial co-state (to a higher degree of accuracy) was found to be

$$\boldsymbol{\lambda}_r(t_0) = \begin{bmatrix} 0.126028606084032 & -0.076645173007368 & -0.056357944401905 \end{bmatrix}^T \tag{30a}$$

$$\boldsymbol{\lambda}_v(t_0) = \begin{bmatrix} 0.039985926336831 & -0.005184908743097 & -0.064101783338762 \end{bmatrix}^T \tag{30b}$$

$$\lambda_m(t_0) = 0.022356555155467 \tag{30c}$$

Reference [4] uses Particle-Swarm Optimization (PSO) to find solutions for the minimum-energy trajectories that can be gradually converted to a minimum-fuel trajectory through continuation. While not being particularly interested in implementing PSO, there is interest in continuation. But given the solution for the minimum-fuel trajectory is already solved for, the only option is to work backwards until the minimum-energy solution is found. This simply involves increasing the value of ϵ from 0 to 1, rather than vice versa. The gradual increases in ϵ were a set of 25 linearly spaced values between 0 and 1. The continuation trajectories are shown in Figure 4 where the color bar represents the value of ϵ with the minimum-fuel being blue and the minimum-energy being red. Figure 5 shows the control profile for the continuation process, which shows how the *bang-bang* control of the minimum-fuel problem is evolved from the minimum-energy solution.

D. The Issue with Initialization

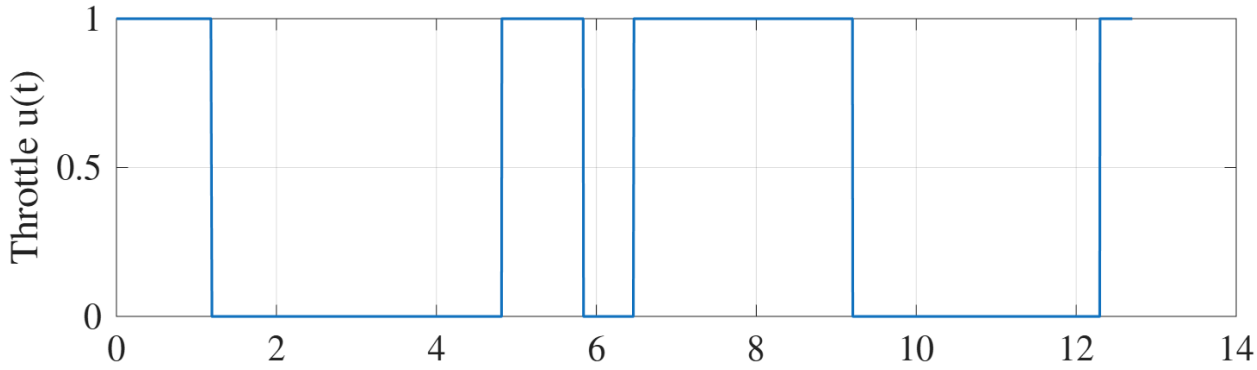
So far this work has looked at recreating results for the L_2 to L_1 halo orbit transfer, and because of this the initial "guesses" of the co-state vector will always result in convergence to an optimal trajectory and control profile. But in a more realistic approach the initial guess will be bad. Reference [4] uses PSO, which is popular, because not only does it help find local minimum, there are additional parts of the algorithm that are exploratory and it can be used to find additional minima in the search for a global solution. That's fairly involved, and I don't feel it meets the point of this write-up as PSO is not in the standard optimal control methodology.

The steepest descent method, however, is quite robust against poor initial guesses. So my idea is to implement steepest descent on the minimum-energy problem, then use those results as the initial guess for the shooting method to achieve a higher accuracy in terminal conditions, and then perform the continuation method to get the minimum-fuel solution.

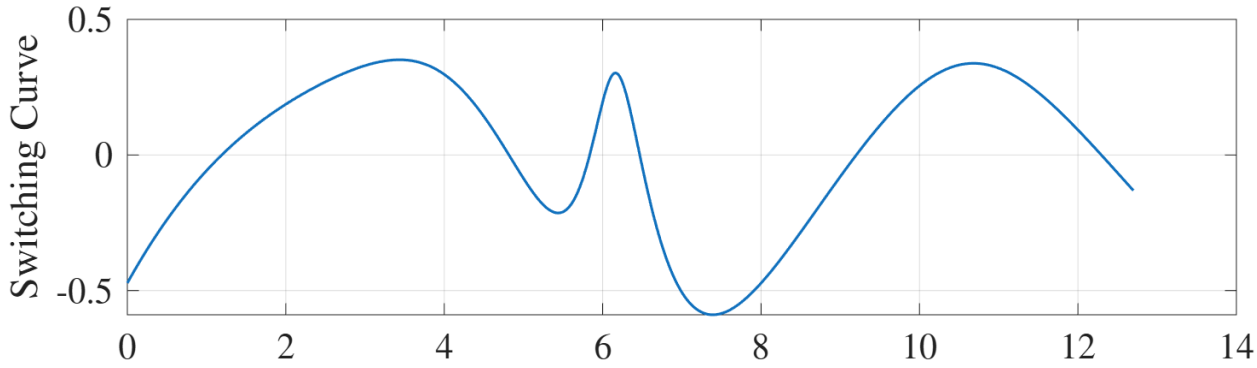
E. Steepest Descent

Graciously, I have received notes from Dr. Nebelecky from when he took Optimal Control with Dr. Singh back in 2008*, and in those notes a procedure for steepest descent with terminal constraints is laid out with credit to Carlos

*Not to make anyone feel terrible, but I was 6 years old in 2008

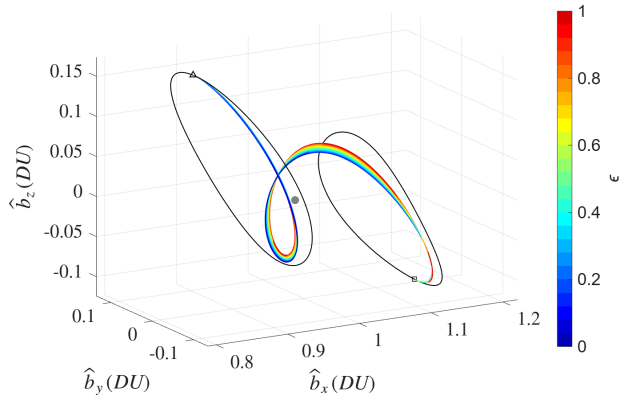


(a) Throttle Factor $u(t)$

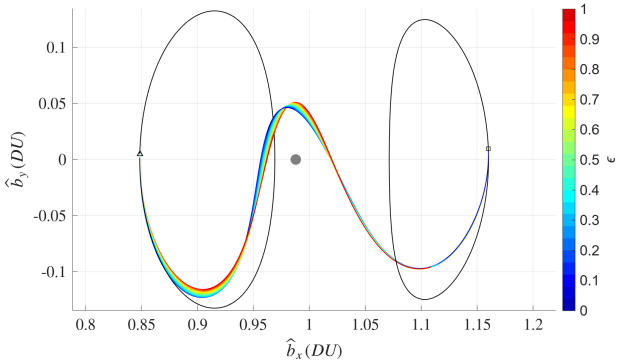


(b) Switching curve

Fig. 3 Minimum-Fuel trajectory control profile

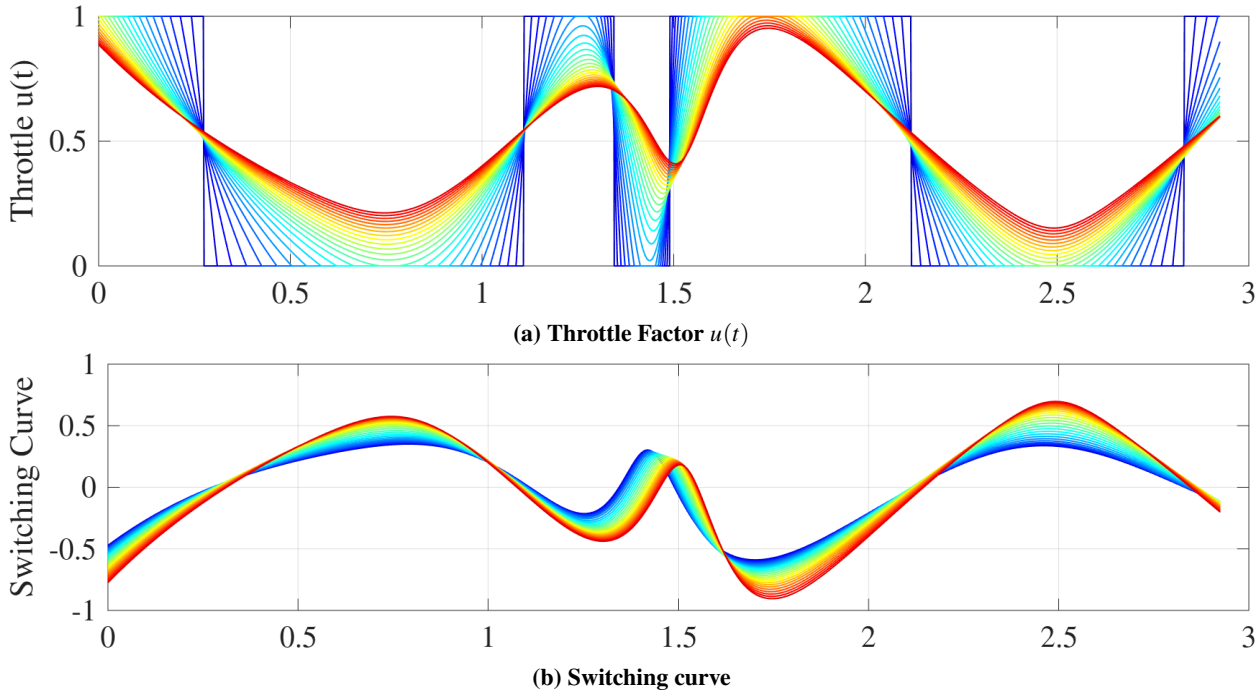


(a) Three-dimensional view



(b) Planar projection

Fig. 4 Minimum-Fuel to minimum-energy continuation trajectories



(a) Throttle Factor $u(t)$

(b) Switching curve

Fig. 5 Minimum-Fuel to minimum-energy continuation control profiles

Roithmayr, 1988. This procedure is more rigorous in enforcing terminal constraints than the suggestion in Reference [6], which simply adds a penalty term to the cost function and makes a slight adjustment to the values of the final co-states.

In general, the method of steepest descent involves guessing the control history for the entire propagation time, using that to propagate the co-state trajectories backwards in time starting from known final values, then propagating the states forward in time using the control profile and co-states. The errors in the propagated states are then used to adjust the new guess for the control profile and this process is repeated until convergence is met.

For the current system, the Roithmayr *Numerical Gradient Procedure for Dynamic Systems with Terminal Constraints* is as follows. The performance index,

$$J_a = \boldsymbol{\phi}(\mathbf{x}(t_f), t_f) + \boldsymbol{\nu}^T \boldsymbol{\psi}(\mathbf{x}(t_f)) + \int_{t_0}^{t_f} \left\{ L + \boldsymbol{\lambda}^T \left[\mathbf{f} - \dot{\mathbf{x}} \right] \right\} \, dt \tag{31}$$

where $\boldsymbol{\phi}$ is a $q \times 1$ vector of terminal constraints, $\boldsymbol{\nu}$ is $q \times 1$ vector of constants akin to Lagrange multipliers on the terminal constraints. As the terminal constraints are only imposed on the position and velocity of the satellite, $q = 6$ For reference, we will denote the size of previously used terms. The state vector and its time derivative, $\mathbf{x}$ and $\dot{\mathbf{x}}$ are $n \times 1$ column vectors, with $\boldsymbol{\lambda}$ being an $n \times 1$ vector of Lagrange multipliers associated with the state dynamics with $n = 7$. The control input vector, $\mathbf{u}$ is formally the $m \times 1$ vector,

$$\mathbf{u} = \begin{bmatrix} u \\ \boldsymbol{\alpha} \end{bmatrix} \tag{32}$$

where $m = 4$. The functions $\boldsymbol{\phi}(\mathbf{x}(t_f), t_f)$ and $\boldsymbol{\psi}(\mathbf{x}(t_f))$ represent a penalty on the terminal conditions and a set of q terminal terminal constraints, respectfully. For this application, these functions are defined as

$$\boldsymbol{\phi}(\mathbf{x}(t_f), t_f) = \mathbf{0}_{n \times 1} \tag{33a}$$

$$\boldsymbol{\psi}(\mathbf{x}(t_f)) = \mathbf{x}_f - \mathbf{x}(t_f) \tag{33b}$$

Finally, the L term is simply the original integrand of the cost function of the minimum-energy problem expressed in Eq.(8). Note we use the minimum-energy formulation in an attempt to avoid the major discontinuities involved when

working with the minimum-fuel problem. After a minimum-energy solution is found, continuation can be used to transition that solution to the minimum-fuel solution (if one exists). We can define the Lagrange multiplier,

$$\boldsymbol{\lambda}^T \equiv \left[\boldsymbol{\lambda}^J\right]^T + \boldsymbol{\nu}^T \left[\boldsymbol{\lambda}^i\right]^T \tag{34}$$

where $\boldsymbol{\lambda}^J$ is an $n \times 1$ vector of Lagrange multipliers of the system that would correspond to the system without any terminal constraints. The $\boldsymbol{\lambda}^i$ term is an $n \times q$ matrix that corresponds to the terminal constraints.The variation in J_a can then be written as,

$$\begin{aligned}\delta J_a = &\left(\frac{\partial \boldsymbol{\phi}}{\partial \mathbf{x}_f} + \boldsymbol{\nu}^T \frac{\partial \boldsymbol{\psi}}{\partial \mathbf{x}_f} - \left[\boldsymbol{\lambda}_f^J\right]^T - \boldsymbol{\nu}^T \left[\boldsymbol{\lambda}^i\right]^T\right) \delta \mathbf{x}_f \\ &+ \int_{t_0}^{t_f} \left\{ \left[\frac{\partial L}{\partial \mathbf{x}} + \left(\left[\boldsymbol{\lambda}^J\right]^T + \boldsymbol{\nu}^T \left[\boldsymbol{\lambda}^i\right]^T\right) \frac{\partial \mathbf{f}}{\partial \mathbf{x}} + \left[\dot{\boldsymbol{\lambda}}^J\right]^T + \boldsymbol{\nu}^T \left[\dot{\boldsymbol{\lambda}}^i\right]^T \right] \delta \mathbf{x} \right. \\ &\left. + \left[\frac{\partial L}{\partial \mathbf{u}} + \left(\left[\boldsymbol{\lambda}^J\right]^T + \boldsymbol{\nu}^T \left[\boldsymbol{\lambda}^i\right]^T\right) \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right] \delta \mathbf{u} \right\} \, dt \end{aligned} \tag{35}$$

where

$$\frac{\partial L}{\partial \mathbf{x}} = \mathbf{0}_{n\times 1} \tag{36a}$$

$$\frac{\partial L}{\partial \mathbf{u}} = \begin{bmatrix} (1 - \epsilon + 2\epsilon u) & 0 & 0 & 0 \end{bmatrix} \tag{36b}$$

and

$$\frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \begin{bmatrix} \mathbf{0} & I & \mathbf{0} \\ G & H & \frac{\lambda_v}{\lambda_v} \frac{u T_{max}}{m^2} \\ \mathbf{0} & \mathbf{0} & 0 \end{bmatrix} \tag{37a}$$

$$\frac{\partial \mathbf{f}}{\partial \mathbf{u}} = \begin{bmatrix} \mathbf{0}_{3\times 1} & \mathbf{0}_{3\times 3} \\ \frac{T_{max}}{c} \boldsymbol{\alpha} & \frac{u T_{max}}{m} I_{3\times 3} \\ -\frac{T_{max}}{c} & \mathbf{0}_{1\times 3} \end{bmatrix} \tag{37b}$$

The differential equations for the multipliers that drive the variation in the state vector, $\delta \mathbf{x}$, to zero are

$$\left[\dot{\boldsymbol{\lambda}}^J\right]^T = -\frac{\partial L}{\partial \mathbf{x}} - \left[\boldsymbol{\lambda}^J\right]^T \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \tag{38a}$$

$$\left[\dot{\boldsymbol{\lambda}}^i\right]^T = -\left[\boldsymbol{\lambda}^i\right]^T \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \tag{38b}$$

These differential equations are propagated backwards with the "initial" conditions at the final time, t_f,

$$\boldsymbol{\lambda}_f^J = \frac{\partial \boldsymbol{\phi}^T}{\partial \mathbf{x}_f} = \mathbf{0}_{n\times 1} \tag{39a}$$

$$\boldsymbol{\lambda}_f^i = \frac{\partial \boldsymbol{\psi}^T}{\partial \mathbf{x}_f} = \begin{bmatrix} I_{q\times q} \\ \mathbf{0}_{1\times q} \end{bmatrix} \tag{39b}$$

Note that the above equations indicate $\boldsymbol{\lambda}^J(t) = \mathbf{0} \quad \forall \; t \in [t_0 \;\; t_f]$. The variation in J_a can now be written as

$$\delta J_a = \delta J + \boldsymbol{\nu}^T \delta \boldsymbol{\psi} \tag{40}$$

where,

$$\delta J = \int_{t_0}^{t_f} \left(\frac{\partial L}{\partial \mathbf{u}} + \left[\boldsymbol{\lambda}^J\right]^T \frac{\partial \mathbf{f}}{\partial \mathbf{u}}\right) \delta \mathbf{u} \, dt \tag{41a}$$

$$\delta \boldsymbol{\psi} = \int_{t_0}^{t_f} \left[\boldsymbol{\lambda}^i\right]^T \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \delta \mathbf{u} \, dt \tag{41b}$$

For notational brevity, we define

$$\Lambda_\phi \equiv \frac{\partial L}{\partial \mathbf{u}} + \left[\boldsymbol{\lambda}^J\right]^T \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \tag{42a}$$

$$\Lambda_\psi \equiv \left[\boldsymbol{\lambda}^i\right]^T \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \delta\mathbf{u} \tag{42b}$$

The variation in control is then chosen to be

$$\delta\mathbf{u} = -K\left[\Lambda_\phi + \boldsymbol{\nu}^T \Lambda_\psi\right]^T \tag{43}$$

where K is a heuristic scalar weighting greater than zero. Substituting the above into Eq. (41b),

$$\delta\boldsymbol{\psi} = -K \int_{t_0}^{t_f} \left[\Lambda_\psi\right]\left[\Lambda_\phi + \boldsymbol{\nu}^T \Lambda_\psi\right]^T \, dt \tag{44}$$

Introducing more shorthand,

$$\mathbf{d} = \int_{t_0}^{t_f} \Lambda_\psi \Lambda_\phi^T \, dt \tag{45a}$$

$$Q = \int_{t_0}^{t_f} \Lambda_\psi \Lambda_\psi^T \, dt \tag{45b}$$

the variation in $\boldsymbol{\psi}$ can be expressed as

$$\delta\boldsymbol{\psi} = -K\left[\mathbf{d} + Q\boldsymbol{\nu}\right] \tag{46}$$

Note that the integration required to obtain the terms in Eq. (45) are evaluated using the trapezoidal integration technique using MATLAB's cumtrapz() functionality. As the end goal is to drive $\delta\boldsymbol{\psi}$ to zero, for a given trajectory we can choose $\delta\boldsymbol{\psi} = -\delta\boldsymbol{\psi}_e$ where $\delta\boldsymbol{\psi}_e$ is simply the value of $\boldsymbol{\psi}(\mathbf{x}(t_f), t_f)$ evaluated at the final time, i.e., the error between the desired final positions and velocities and the propagated values for a given iteration of the algorithm. Setting the value of $\delta\boldsymbol{\psi}$ lets us solve for $\boldsymbol{\nu}$.

$$\boldsymbol{\nu} = -Q^{-1}\left[\mathbf{d} - \frac{1}{K}\delta\boldsymbol{\psi}_e\right] \tag{47}$$

The value of $\boldsymbol{\nu}$ lets us solve for the variation in the control vector,

$$\delta\mathbf{u} = -K\left[\Lambda_\phi + \boldsymbol{\nu}^T \Lambda_\psi\right]^T \tag{48}$$

and finally we can update the initially guessed control vector with

$$\mathbf{u}_{\text{new}} = \mathbf{u}_{\text{old}} + \delta\mathbf{u} \tag{49}$$

This process is repeated until one of two sets of convergence criteria is met. The first being the norm of error between states is less than a tolerance of $\gamma_1 = 1 \times 10^{-4}$, or if the difference between consecutive cost function evaluations is less than a tolerance of $\gamma_2 = 1 \times 10^{-5}$. In other words, the termination criteria is

$$h_{\text{norm}} = \|\mathbf{x}_f - \mathbf{x}(t_f)\| < \gamma_1 \tag{50}$$

$$J_{\text{norm}} = |J_{k+1} - J_k| < \gamma_2 \tag{51}$$

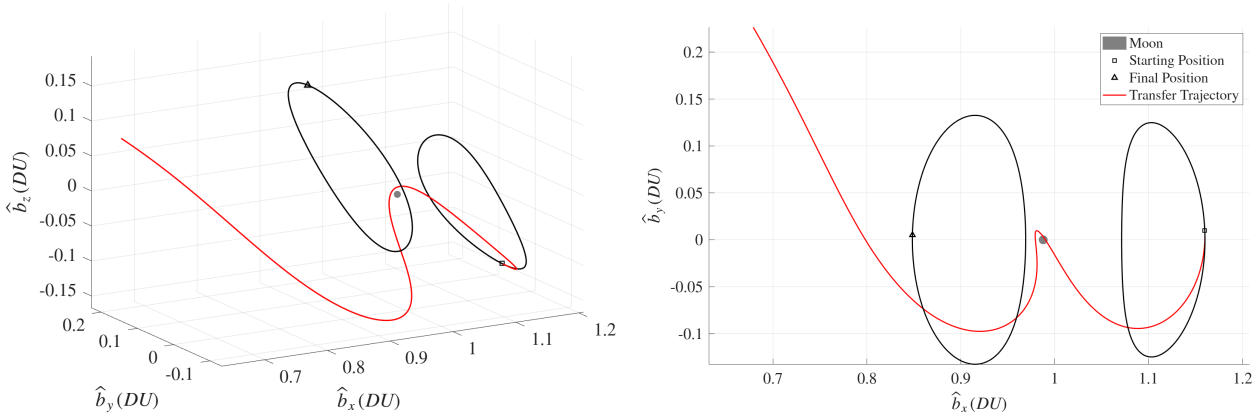
III. Results

A. L_2 to L_1 Transfer

Once again, we first look to confirm the choice of methodology by solving for the same fuel optimal trajectory of the L_2 to L_1 transfer from Section II.C. The initial guess to initialize the steepest descent algorithm is,

$$u_0(t) = 0.5 \tag{52a}$$

$$\boldsymbol{\alpha}_0(t) = \begin{bmatrix} -1 & 0 & 0 \end{bmatrix}^T \tag{52b}$$



(a) Three-dimensional view **(b) Planar projection**

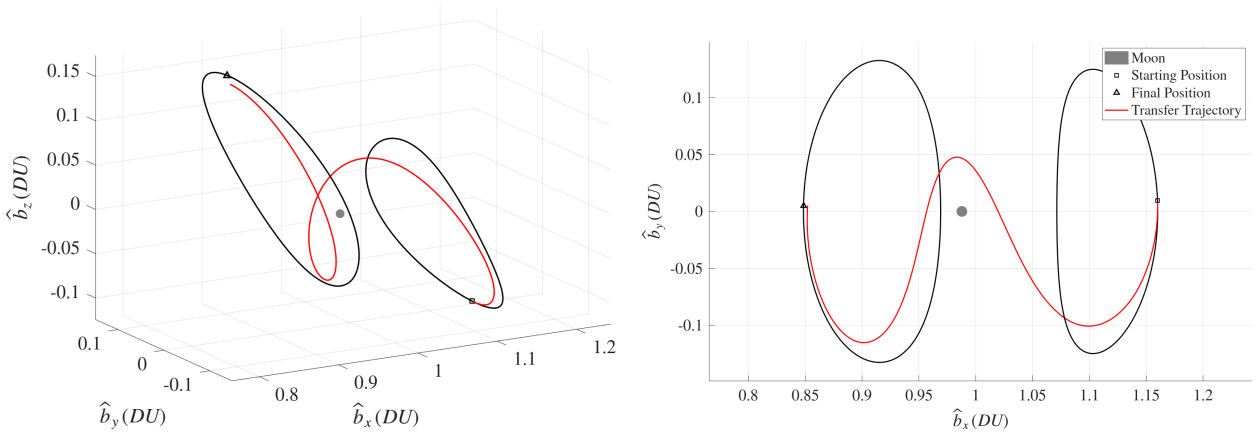
Fig. 6 Initial guess trajectory

which implies the satellite is thrust at half of its maximum thrust capability and for the entire transfer is thrusting in the negative $\mathbf{e}_x$ direction of the synodic frame. The initial transfer is displayed in Figure 6, and is obviously a poor guess, with the final position being over 10,000 km away from the desired final position The heuristic weighting introduced in Eq. (43) is set to $K = 5$ (after some trial and error) and after 59 iterations the steepest descent algorithm ends. The final trajectory from the steepest descent algorithm is displayed in Figure 7. The initial co-state vector from the final iteration is

$$\boldsymbol{\lambda}_r(t_0) = \begin{bmatrix} 0.136497269706572 & -0.063883194523382 & -0.071522759789855 \end{bmatrix}^T \tag{53a}$$

$$\boldsymbol{\lambda}_v(t_0) = \begin{bmatrix} 0.044363376274060 & -0.006167976512232 & -0.074759406273468 \end{bmatrix}^T \tag{53b}$$

$$\lambda_m(t_0) = 0.023120214589177 \tag{53c}$$

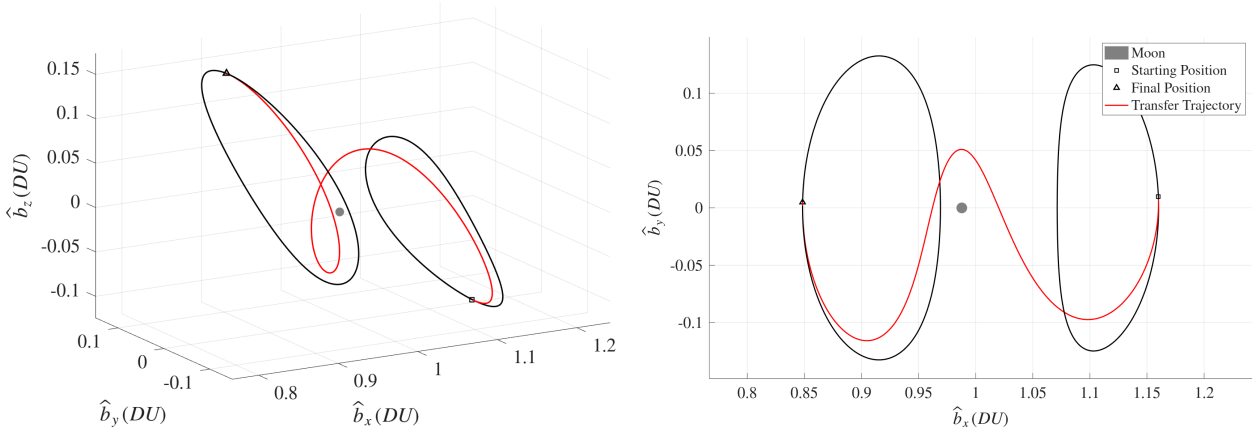


(a) Three-dimensional view **(b) Planar projection**

Fig. 7 Final trajectory using steepest descent

The final position of the steepest descent algorithm is still about 4,300 km away from the desired final position. This is actually terrible convergence, and different values of K were able to reduce the error in final position to just hundreds of kilometers, but the transition from those solutions to the shooting method did not work as the steepest descent control profile had too many discontinuities after adjustments to enforce the control constraints were imposed. The co-state vector in Eq. (53) is used to initialize the the shooting method algorithm for refinement. The shooting algorithm was

able to converge in just 7 iterations using the results from the steepest descent work. The trajectory is shown in Figure 8 and the throttle control profile and switching curve is displayed in Figure 9.

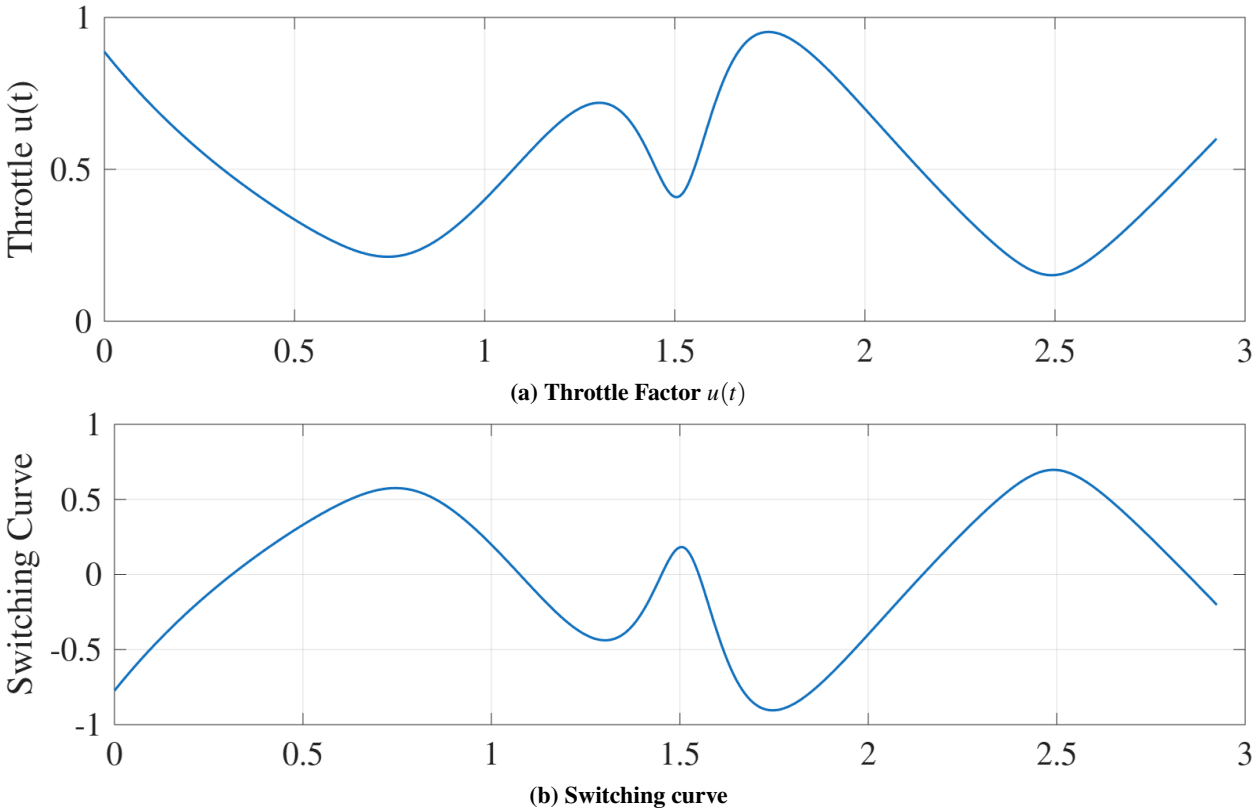


(a) Three-dimensional view **(b) Planar projection**

Fig. 8 Final trajectory using shooting method

This correlates to the minimum-energy (i.e., $\epsilon = 1$) solution shown in Figures 4 and 5. The minimum-fuel optimal trajectory shown in Figure 2 can be obtained using the continuation method as before, but transitioning ϵ from 1 to 0.

This solution is acceptable, and other initial guesses were able to converge to the same trajectory. These other



(a) Throttle Factor $u(t)$

(b) Switching curve

Fig. 9 Control profile using shooting method

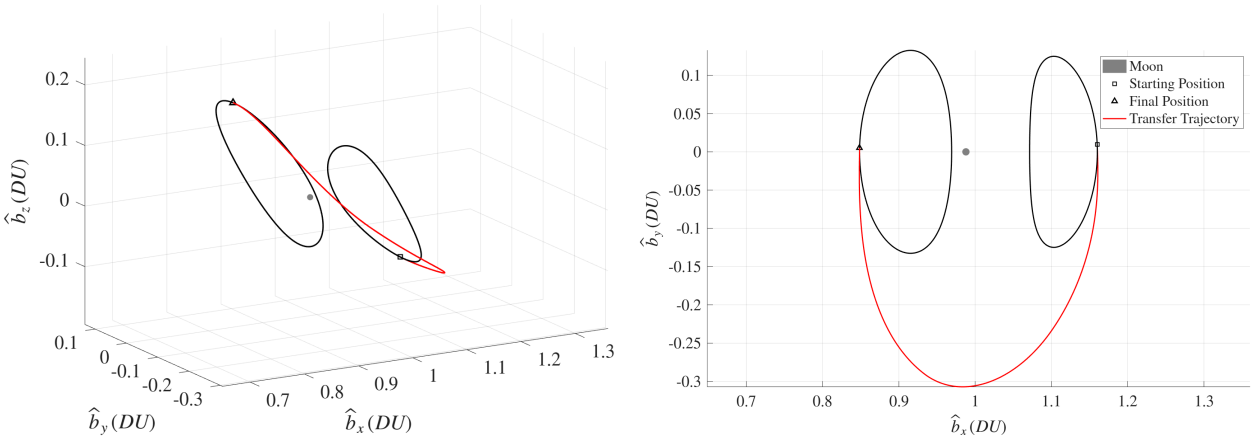
guesses include,

$$u_{0,2}(t) = 0.75 \qquad u_{0,3}(t) = 0.5\cos(\pi t) + 0.5$$
$$\boldsymbol{\alpha}_{0,2}(t) = \begin{bmatrix} 0 & -1 & 0 \end{bmatrix}^T \qquad \boldsymbol{\alpha}_{0,3}(t) = \text{norm}\left(\begin{bmatrix} 0.1\sin(2\pi t) & \cos(2\pi t) & \sin(2\pi t + \pi/16) \end{bmatrix}^T\right)$$

where norm() indicates the vector is normalized to a unit vector for all times. However not all roads lead to this same trajectory. An initial guess of

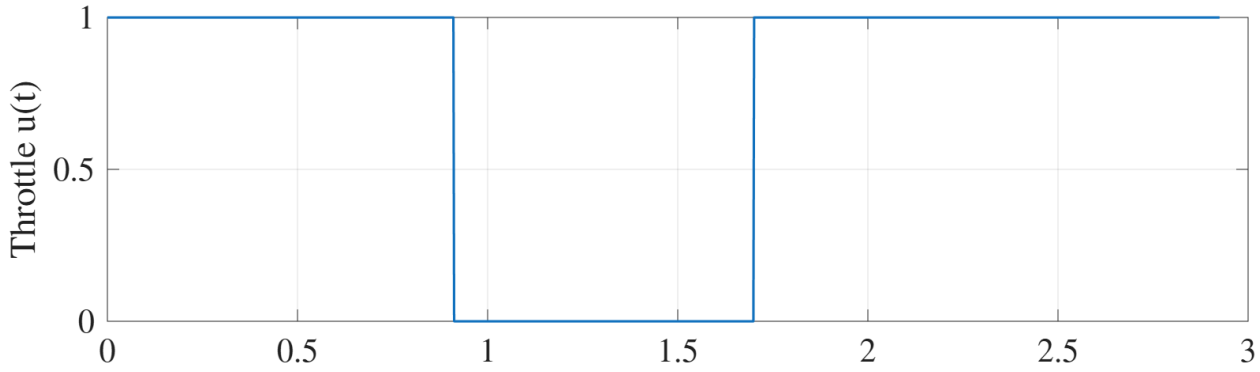
$$u_0(t) = .75 \tag{54a}$$
$$\boldsymbol{\alpha}_0(t) = \begin{bmatrix} -1 & 0 & 0 \end{bmatrix}^T \tag{54b}$$

leads to a completely different trajectory. After a complete transition from steepest descent to the shooting method as well the continuation problem converting the minimum-energy solution to minimum-fuel, the trajectory and control profiles are shown in Figures 10 and 11, respectfully. While the mathematical conditions for optimality are met regarding the formulation of the problem, the final change in mass is $\Delta m = 61.27$ kg, so this is clearly a local minima as opposed to a global solution. This trajectory was also found in Ref [4] using the PSO methods.

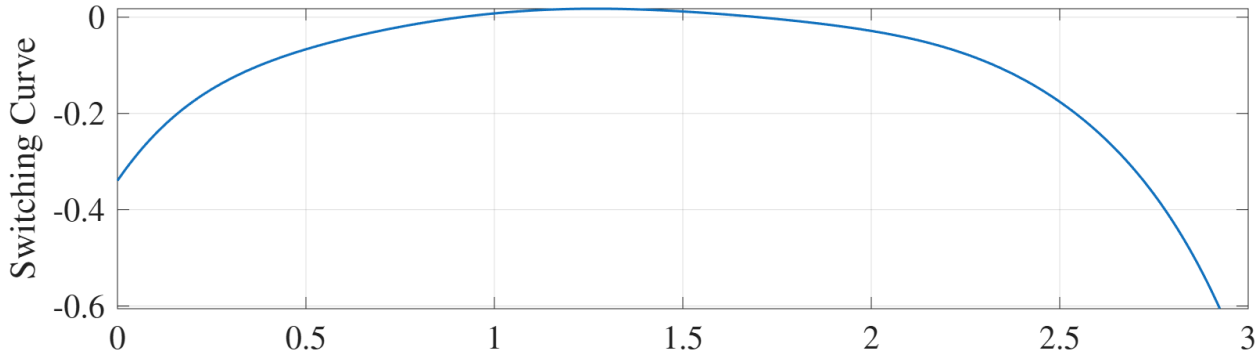


(a) Three-dimensional view **(b) Planar projection**

Fig. 10 Alternative optimal trajectory



(a) Throttle Factor $u(t)$



(b) Switching curve

Fig. 11 Alternative optimal control profile

B. L_1 to L_2 Transfer

In an effort to find a trajectory that wasn't explicitly solved in another work I thought it would be neat to do the reverse trajectory, where the satellite must now traverse from the L_1 halo orbit to the L_2 halo orbit. The mirror image of the original problem isn't terribly different but the equations of motion under CR3BP are not symmetrical across the $\mathbf{e}_y$ axis because the affect of the rotating synodic frame is felt. The initial and final conditions are simply switched, because transfers from perilune to perilune are more stable than apolune to apolune transfers, as was frustratingly discovered. The boundary conditions, more formally, are

$$\mathbf{r}_0 = \begin{bmatrix} 0.8484736688482315 & 0.00506488863463682 & 0.17343680487577373 \end{bmatrix}^T \; DU \tag{55a}$$

$$\mathbf{v}_0 = \begin{bmatrix} 0.005241131023638693 & 0.26343491250951045 & -0.008541420325316247 \end{bmatrix}^T \; VU \tag{55b}$$

$$\mathbf{r}_f = \begin{bmatrix} 1.1599795702248494 & 0.009720428035815552 & -0.12401864915284157 \end{bmatrix}^T \; DU \tag{55c}$$

$$\mathbf{v}_f = \begin{bmatrix} 0.008477705130550553 & -0.20786307954141953 & -0.010841912833115475 \end{bmatrix}^T \; VU \tag{55d}$$

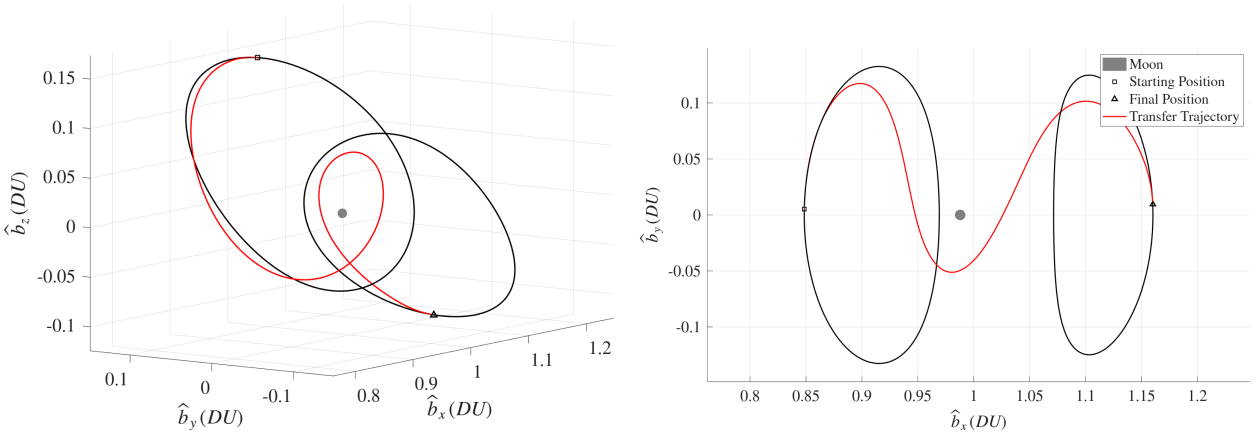
$$\tag{55e}$$

The initial guess for the control profile are

$$u_0(t) = 0.5 \tag{56a}$$

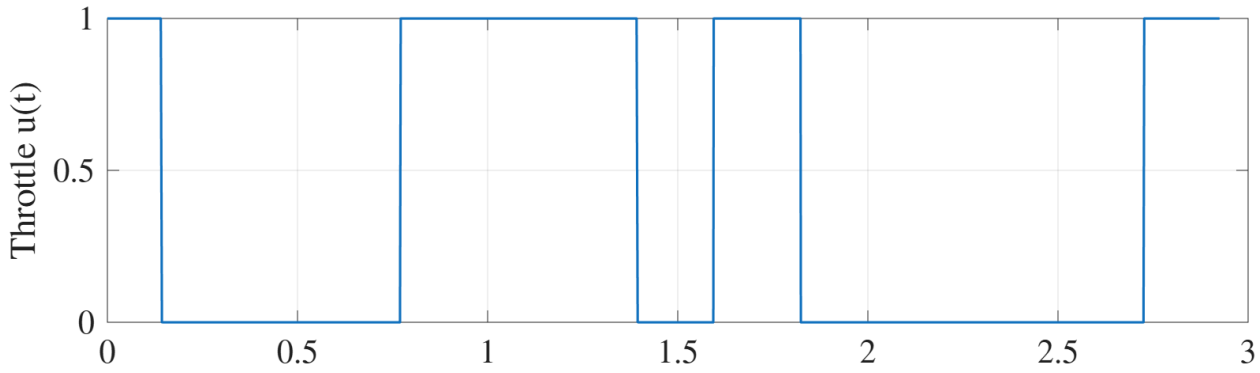
$$\boldsymbol{\alpha}_0(t) = \begin{bmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \end{bmatrix}^T \tag{56b}$$

and the weighting factor for the steepest descent algorithm is set to $K = 10$ to ensure a proper conversion to the shooting method process. The full process from the steepest descent initialization to the continuation process to arrive at the minimum-fuel solution is found in Figures 12 and 13. We see the trajectory mirrors that of the L_2 to L_1 transfer about the $\mathbf{e}_x$ axis, which is not unusual given the symmetric nature of the CR3BP. The final change in mass was calculated to be $\Delta m = 34.23$ kg which is slightly less than the original transfer, but close enough to confirm the similarities in the two transfers.

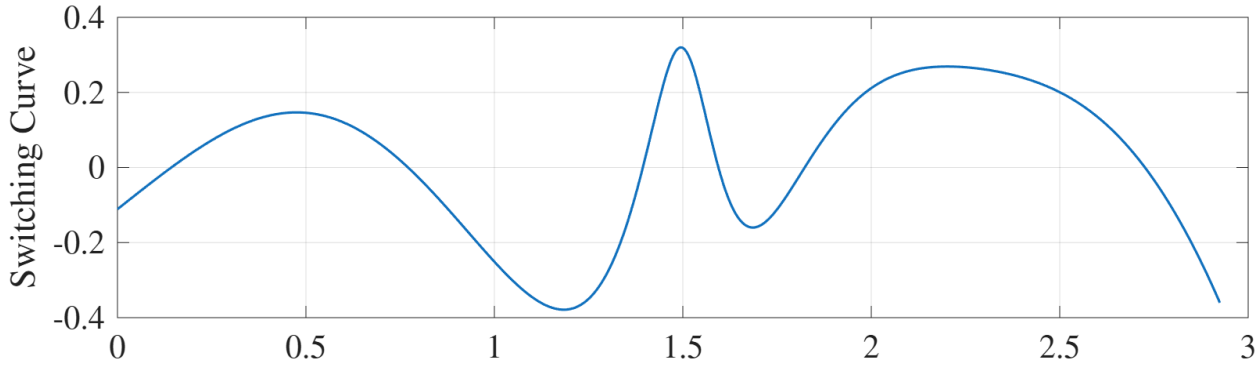


(a) Three-dimensional view **(b) Planar projection**

Fig. 12 L_1 to L_2 minimum-fuel trajectory



(a) Throttle Factor $u(t)$



(b) Switching curve

Fig. 13 L_1 to L_2 minimum-fuel control profile

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A. Extended Derivatives

In order to compute the matrices defined in Eqs. (24) and (25) there is a smaller, but no less important Jacobian needed. The expression,

$$\frac{\partial \left(G^T \boldsymbol{\lambda}_v\right)}{\partial \mathbf{r}} \tag{A.1}$$

will be denoted matrix A and can be rewritten as,

$$A = \frac{\partial}{\partial \mathbf{r}} \left(\begin{bmatrix} G_{11} & G_{12} & G_{13} \\ G_{21} & G_{22} & G_{23} \\ G_{31} & G_{32} & G_{33} \end{bmatrix} \boldsymbol{\lambda}_v \right) = \frac{\partial}{\partial \mathbf{r}} \left(\begin{bmatrix} G_{11}\lambda_{v,1} + G_{12}\lambda_{v,2} + G_{13}\lambda_{v,3} \\ G_{21}\lambda_{v,1} + G_{22}\lambda_{v,2} + G_{23}\lambda_{v,3} \\ G_{31}\lambda_{v,1} + G_{32}\lambda_{v,2} + G_{33}\lambda_{v,3} \end{bmatrix} \right) \tag{A.2}$$

Luckily, the $\boldsymbol{\lambda}_v$ term is not an explicit function of $\mathbf{r}$ and we can simply take the derivatives of terms regarding the G matrix. The necessary partial derivatives are

$$\frac{\partial G_{11}}{\partial x} = \frac{9(1-\mu)(x+\mu)}{r_1^5} - \frac{15(1-\mu)(x+\mu)^3}{r_1^7} + \frac{9\mu(x+\mu-1)}{r_2^5} - \frac{15\mu(x+\mu-1)^3}{r_1^7} \tag{A.3a}$$

$$\frac{\partial G_{11}}{\partial y} = \frac{3(1-\mu)y}{r_1^5} - \frac{15(1-\mu)(x+\mu)^2 y}{r_1^7} + \frac{3\mu y}{r_2^5} - \frac{15\mu(x+\mu-1)^2 y}{r_2^7} \tag{A.3b}$$

$$\frac{\partial G_{11}}{\partial z} = \frac{3(1-\mu)z}{r_1^5} - \frac{15(1-\mu)(x+\mu)^2 z}{r_1^7} + \frac{3\mu z}{r_2^5} - \frac{15\mu(x+\mu-1)^2 z}{r_2^7} \tag{A.3c}$$

$$\frac{\partial G_{22}}{\partial x} = \frac{3(1-\mu)(x+\mu)}{r_1^5} - \frac{15(1-\mu)y^2(x+\mu)}{r_1^7} + \frac{3\mu(x+\mu-1)}{r_2^5} - \frac{15\mu y^2(x+\mu-1)}{r_2^7} \tag{A.4a}$$

$$\frac{\partial G_{22}}{\partial y} = \frac{9(1-\mu)y}{r_1^5} - \frac{15(1-\mu)y^3}{r_1^7} + \frac{9\mu y}{r_2^5} - \frac{15\mu y^3}{r_2^7} \tag{A.4b}$$

$$\frac{\partial G_{22}}{\partial z} = \frac{3(1-\mu)z}{r_1^5} - \frac{15(1-\mu)y^2 z}{r_1^7} + \frac{3\mu z}{r_2^5} - \frac{15\mu y^2 z}{r_2^7} \tag{A.4c}$$

$$\frac{\partial G_{33}}{\partial x} = \frac{3(1-\mu)(x+\mu)}{r_1^5} - \frac{15(1-\mu)z^2(x+\mu)}{r_1^7} + \frac{3\mu(x+\mu-1)}{r_2^5} - \frac{15\mu z^2(x+\mu-1)}{r_2^7} \tag{A.5a}$$

$$\frac{\partial G_{33}}{\partial y} = \frac{3(1-\mu)y}{r_1^5} - \frac{15(1-\mu)z^2 y}{r_1^7} + \frac{3\mu y}{r_2^5} - \frac{15\mu z^2 y}{r_2^7} \tag{A.5b}$$

$$\frac{\partial G_{33}}{\partial z} = \frac{9(1-\mu)z}{r_1^5} - \frac{15(1-\mu)z^3}{r_1^7} + \frac{9\mu z}{r_2^5} - \frac{15\mu z^3}{r_2^7} \tag{A.5c}$$

$$\frac{\partial G_{12}}{\partial x} = \frac{3(1-\mu)y}{r_1^5} - \frac{15(1-\mu)y(x+\mu)^2}{r_1^7} + \frac{3\mu y}{r_2^5} - \frac{15\mu y(x+\mu-1)^2}{r_2^7} \tag{A.6a}$$

$$\frac{\partial G_{12}}{\partial y} = \frac{3(1-\mu)(x+\mu)}{r_1^5} - \frac{15(1-\mu)(x+\mu)y^2}{r_1^7} + \frac{3\mu(x+\mu-1)}{r_2^5} - \frac{15\mu * (x+\mu-1)y^2}{r_2^7} \tag{A.6b}$$

$$\frac{\partial G_{12}}{\partial z} = -\frac{15(1-\mu)(x+\mu)yz}{r_1^7} - \frac{15\mu(x+\mu-1)yz}{r_2^7} \tag{A.6c}$$

$$\frac{\partial G_{13}}{\partial x} = \frac{3(1-\mu)z}{r_1^5} - \frac{15(1-\mu)z(x+\mu)^2}{r_1^7} + \frac{3\mu z}{r_2^5} - \frac{15\mu z(x+\mu-1)^2}{r_2^7} \tag{A.7a}$$

$$\frac{\partial G_{13}}{\partial y} = \frac{\partial G_{12}}{\partial z} \tag{A.7b}$$

$$\frac{\partial G_{13}}{\partial z} = \frac{3(1-\mu)(x+\mu)}{r_1^5} - \frac{15(1-\mu)(x+\mu)z^2}{r_1^7} + \frac{3\mu(x+\mu-1)}{r_2^5} - \frac{15\mu(x+\mu-1)z^2}{r_2^7} \tag{A.7c}$$

$$\frac{\partial G_{23}}{\partial x} = -\frac{15(1-\mu)yz(x+\mu)}{r_1^7} - \frac{15\mu yz(x+\mu-1)}{r_2^7} \tag{A.8a}$$

$$\frac{\partial G_{23}}{\partial y} = \frac{3(1-\mu)z}{r_1^5} - \frac{15(1-\mu)zy^2}{r1^7} + \frac{3\mu z}{r_2^5} - \frac{15\mu zy^2}{r_2^7} \tag{A.8b}$$

$$\frac{\partial G_{23}}{\partial z} = \frac{3(1-\mu)y}{r_1^5} - \frac{15(1-\mu)yz^2}{r_1^7} + \frac{3\mu y}{r_2^5} - \frac{15\mu yz^2}{r_2^7} \tag{A.8c}$$

The matrix A can also be written as

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \tag{A.9}$$

where

$$a_{11} = \begin{bmatrix} \frac{\partial G_{11}}{\partial x} & \frac{\partial G_{12}}{\partial x} & \frac{\partial G_{13}}{\partial x} \end{bmatrix} \boldsymbol{\lambda}_v \tag{A.10a}$$

$$a_{12} = \begin{bmatrix} \frac{\partial G_{11}}{\partial y} & \frac{\partial G_{12}}{\partial y} & \frac{\partial G_{13}}{\partial y} \end{bmatrix} \boldsymbol{\lambda}_v \tag{A.10b}$$

$$a_{13} = \begin{bmatrix} \frac{\partial G_{11}}{\partial z} & \frac{\partial G_{12}}{\partial z} & \frac{\partial G_{13}}{\partial z} \end{bmatrix} \boldsymbol{\lambda}_v \tag{A.10c}$$

$$a_{21} = \begin{bmatrix} \frac{\partial G_{12}}{\partial x} & \frac{\partial G_{22}}{\partial x} & \frac{\partial G_{23}}{\partial x} \end{bmatrix} \boldsymbol{\lambda}_v \tag{A.10d}$$

$$a_{22} = \begin{bmatrix} \frac{\partial G_{12}}{\partial y} & \frac{\partial G_{22}}{\partial y} & \frac{\partial G_{23}}{\partial y} \end{bmatrix} \boldsymbol{\lambda}_v \tag{A.10e}$$

$$a_{23} = \begin{bmatrix} \frac{\partial G_{12}}{\partial z} & \frac{\partial G_{22}}{\partial z} & \frac{\partial G_{23}}{\partial z} \end{bmatrix} \boldsymbol{\lambda}_v \tag{A.10f}$$

$$a_{31} = \begin{bmatrix} \frac{\partial G_{13}}{\partial x} & \frac{\partial G_{23}}{\partial x} & \frac{\partial G_{33}}{\partial x} \end{bmatrix} \boldsymbol{\lambda}_v \tag{A.10g}$$

$$a_{32} = \begin{bmatrix} \frac{\partial G_{13}}{\partial y} & \frac{\partial G_{23}}{\partial y} & \frac{\partial G_{33}}{\partial y} \end{bmatrix} \boldsymbol{\lambda}_v \tag{A.10h}$$

$$a_{33} = \begin{bmatrix} \frac{\partial G_{13}}{\partial z} & \frac{\partial G_{23}}{\partial z} & \frac{\partial G_{33}}{\partial z} \end{bmatrix} \boldsymbol{\lambda}_v \tag{A.10i}$$

$$\tag{A.10j}$$